PADEE.AI

Product Requirements Document

Version 4.2 -- Quality Control + Agent Architecture + Open Source Model Strategy

v4.2 | April 2026 | Confidential

v4.0	**v4.1**	**v4.2 (this)**
RAG + LLM-agnostic + Cost model	Class 8-12 all subjects + Pricing + Break-even	Quality control + Agent architecture + OSS models

What Changed from v4.1 to v4.2

Three architectural decisions added. The core product, RAG architecture, LLM routing, and roadmap from v4.1 are unchanged.

NEW IN v4.2

Section 6 -- AI Quality Control Framework

Six-layer quality control pipeline added. Layer 1-3 are preventive (RAG, structured prompts, memory context). Layer 4-5 are reactive feedback loops (student ratings, teacher review queue). Layer 6 is an automated evaluation agent (Phase 2) that scores every response before showing it. Quality metrics defined. Fine-tuning quality data pipeline added for Phase 3.

NEW IN v4.2

Section 7 -- Agent Architecture

Formal decision framework for when to use agents vs simple LLM calls. Three agentic patterns defined (tool use, multi-step, self-check). Task-to-approach mapping for all 8 Padee task types. One agent addition for Phase 1: worksheet validation loop. JEE/NEET complex solver agent scoped to Phase 3.

CHANGED IN v4.2

Section 5 -- LLM strategy updated with open source model analysis

Before: LiteLLM routing table without comparative OSS analysis or self-hosting guidance

After: Full open source model comparison table with costs and quality ratings. Self-hosting break-even analysis (not viable until 1M+ students). Fine-tuning roadmap: fine-tune Llama-3.1-8B on student data at Phase 3 for domain-specific quality at 10x lower cost. Photo doubt model changed from GPT-4o-mini to Llama-3.2-11B Vision (40% cheaper).

1. Product Overview

Padee.ai is an AI-first K12 learning platform for India. CBSE Classes 8-12 all subjects from Phase 1, powered by RAG over uploaded NCERT content. LLM-agnostic via LiteLLM. Cost target: Rs.500/student/year. AI quality controlled through six layers. Agents used selectively where multi-step reasoning justifies cost.

Product	Domain	Phase 1 scope	Cost ceiling
Padee.ai	padee.ai (live)	CBSE Class 8-12, all subjects	Rs.500/student/year

2. Complete Feature List (updated in v4.2)

Full feature list from v4.1 is unchanged except for the additions below. New features marked with a star are added in v4.2.

2A. Core AI + Quality Control (updated)

Feature	Description	Phase	Priority
Ask AI -- text doubt	RAG-grounded streaming response via dual-loop solver. Grounded in NCERT. Student memory injected.	1	**P0**
Ask AI -- photo doubt	Llama-3.2-11B Vision (replaces GPT-4o-mini) -- 40% cheaper, comparable quality for textbook images	1	**P0**
Thumbs up/down signal *	After every AI response. Logged to PostHog. Drives helpful_rate metric per subject and chapter.	1	**P0**
Report incorrect answer *	One-tap flag on any AI response. Goes to teacher review queue. Triggers prompt audit if rate exceeds 5%.	1	**P0**
Teacher review queue *	Simple /admin/flagged panel. Teacher marks response correct/wrong. Corrections become labelled training data.	1	**P0**
Quality metrics dashboard *	helpful_rate, flag_rate, regeneration_rate, correction_rate tracked in PostHog. Alert thresholds defined.	1	**P0**
Worksheet validation agent *	Generate questions -> validate each for NCERT accuracy (Llama-8B check) -> regenerate failures. Phase 1 agent addition.	1	**P0**
Automated eval agent	Llama-8B scores every doubt response before showing: NCERT-aligned? Addresses question? Keywords present?	2	**P0**
Fine-tuning pipeline	Collect quality-labelled responses from teacher review. Fine-tune Llama-3.1-8B on student data at Phase 3.	3	**P0**
Custom fine-tuned model	Fine-tuned Llama-8B replaces 80% of API calls at 10x lower cost with higher CBSE-specific accuracy	3	**P1**

2B. Platform & Infrastructure (updated model routing)

Feature	Description	Phase	Priority
LiteLLM abstraction	Single interface to all providers. Model per task from .env. Zero code changes to switch.	1	**P0**
Model routing by task *	Text doubts: Llama-70B. Photo doubts: Llama-3.2-11B Vision. Validation: Llama-8B. Worksheet: GPT-4o-mini.	1	**P0**
Semantic cache (pgvector)	Cache AI responses by question embedding. 50%+ cost reduction on repeated doubts.	1	**P0**
NCERT RAG store	All NCERT PDFs indexed in Supabase pgvector. Admin upload panel at padee.ai/admin.	1	**P0**
Daily usage cap	10 doubts/day free tier. Hard cap server-side for cost control.	1	**P0**
Response length limits	max_tokens per task type. 500 simple doubts, 800 complex, 2000 worksheets.	1	**P0**

3. DeepTutor Feature Incorporation (unchanged from v4.1)

Four features from DeepTutor (HKUDS/DeepTutor, 10.3k GitHub stars). All four in Phase 1.

Feature	Padee implementation	Effort
Dual-loop solver	Analyse loop -> Solve loop. Two sequential LLM calls. Targets the specific misconception, not the general topic.	2 days
Visual explain chip	Explain visually action chip -> LLM generates self-contained HTML with inline SVG. Sandboxed iframe in chat.	3 days
CBSE Paper Mimic	Upload board exam PDF -> text extracted -> LLM clones question style and mark distribution.	3 days
Persistent memory	studentMemory.ts reads Supabase, formats profile, injected into every LLM prompt. 1 file + 3 lines per route.	1 day

4. RAG Architecture -- NCERT-Grounded Answers (unchanged from v4.1)

Every AI answer grounded in uploaded NCERT content -- not LLM training memory. Student asks doubt -> question embedded -> NCERT chunks retrieved from pgvector -> injected into prompt. Syllabus updates: re-upload one PDF.

Step	What happens
1 -- Student asks	Types question or uploads photo of textbook page
2 -- Check cache	Embed question -> search response_cache for 92%+ similarity match -> return cached if found (no LLM call)
3 -- Retrieve NCERT	Embed question -> cosine search ncert_chunks table -> top 4 passages by subject + class filter
4 -- Inject context	System prompt + student memory + NCERT passages + question. Instruction: Answer using ONLY this NCERT content.
5 -- Stream response	LLM streams response word by word. Cached on completion. XP awarded. Doubt saved to DB.
6 -- Quality signal	Student sees thumbs up/down. Flag button. Rating feeds helpful_rate metric.

NCERT corpus: ~48 PDFs, ~10,200 pages, ~5M tokens. One-time embedding cost: approximately Rs.8 ($0.10). Syllabus update: re-upload changed PDF, old chunks replaced automatically.

5. LLM-Agnostic Architecture + Open Source Model Strategy

LiteLLM provides a single interface to all LLM providers. Model selection is per-task via environment variables. No code changes to switch any model.

5A. Open source model comparison (updated in v4.2)

All models below are open source or open-weight. Costs shown are API prices -- you are not paying for the model, only for compute.

Model	Provider	Cost per 1K tokens	Padee task	Quality vs GPT-4o
Llama-3.1-70B	Groq	Rs.0.049 in / Rs.0.066 out	Simple CBSE doubts, practice gen	90% for CBSE tasks
Llama-3.1-8B	Groq	Rs.0.004 in / Rs.0.007 out	Home curation, validation, JSON tasks	80% for structured tasks
Llama-3.2-11B Vision *	Groq	Rs.0.015 in / Rs.0.015 out	Photo doubts -- replaces GPT-4o-mini	85% for textbook images
Qwen-2.5-72B	Together.ai	Rs.0.034 in / Rs.0.034 out	Hindi AI responses (Phase 4)	92% for Hindi tasks
Gemma-2-9B	Groq	Rs.0.017 in / Rs.0.017 out	Agent validation steps, scoring	82% for classification
Fine-tuned Llama-8B *	Self-hosted (Phase 3)	Fixed server cost only	All CBSE doubts post-50K dataset	Can exceed GPT-4o on CBSE tasks
GPT-4o-mini	OpenAI	Rs.0.013 in / Rs.0.050 out	Worksheet gen, evaluation fallback	Baseline 100%
Claude Sonnet	Anthropic	Rs.0.252 in / Rs.1.26 out	JEE advanced solving (Phase 3 only)	105% -- best reasoning

5B. Updated model routing (.env config)

Photo doubt model changed from GPT-4o-mini to Llama-3.2-11B Vision. 40% cheaper on output tokens. Comparable quality for reading printed textbook questions.

```
# Padee.ai LLM routing -- all task assignments configurable

LLM_DOUBT_SIMPLE=groq/llama-3.1-70b-versatile
LLM_DOUBT_VISION=groq/llama-3.2-11b-vision-preview   # CHANGED from gpt-4o-mini
LLM_DOUBT_COMPLEX=claude-sonnet-4-20250514            # Phase 3 only
LLM_PRACTICE_GEN=groq/llama-3.1-8b-instant
LLM_EVALUATION=gpt-4o-mini
LLM_WORKSHEET=gpt-4o-mini
LLM_HOME_CURATOR=groq/llama-3.1-8b-instant
LLM_VISUAL_EXPLAIN=gpt-4o-mini
LLM_VALIDATION=groq/llama-3.1-8b-instant             # NEW -- worksheet validation agent
LLM_QUALITY_EVAL=groq/gemma2-9b-it                   # Phase 2 -- automated quality gate
```

```
LLM_HINDI=together_ai/Qwen/Qwen2.5-72B-Instruct      # Phase 4

EMBEDDING_MODEL=text-embedding-3-small
LLM_FALLBACK=gpt-4o-mini
```

5C. Self-hosting analysis -- when does it make sense?

Self-hosting is not economically viable until approximately 1 million students. Cloud API providers benefit from GPU pools at a scale Padee cannot match. Use API providers until then.

Server configuration	Monthly cost	Model capacity	Break-even student count
AWS g4dn.xlarge (1x T4 GPU)	Rs.44,000	Llama-8B, 5,000 req/hour	1.2 million students
AWS g4dn.4xlarge (2x T4)	Rs.1.4 lakh	Llama-70B, 2,000 req/hour	3.7 million students
AWS p3.2xlarge (V100)	Rs.2.6 lakh	Llama-70B, 3,000 req/hour	6.8 million students

At 50,000 students, API costs via Groq are approximately Rs.15,000/month. A g4dn.xlarge server costs Rs.44,000/month. API access is 3x cheaper at this scale. Self-hosting only makes sense beyond 1 million students.

5D. Fine-tuning roadmap -- Phase 3 (the real open source play)

Fine-tuning a small open source model on proprietary student data is the most important open source decision Padee will make. It converts the data flywheel into a model quality advantage.

Phase	Action	Outcome
Phase 1 (now)	Collect all doubt sessions + teacher corrections in DB. Tag each response as helpful/correct/wrong from student and teacher signals.	Builds the labelled training dataset. Target: 10,000+ labelled pairs by Month 6.
Phase 2 (M5-9)	Clean and structure the dataset. Remove low-quality responses. Create CBSE-specific instruction-following pairs.	Clean training dataset ready. 20,000+ high-quality pairs.
Phase 3 (M10-15)	Fine-tune Llama-3.1-8B on the dataset using QLoRA. Estimated cost: $500 on cloud GPU. Deploy via Together.ai or Modal.	Custom Padee model outperforms base Llama-70B on CBSE tasks. Cost: Rs.0.003/1K tokens (10x cheaper than Groq 70B).
Phase 3+	Iterative fine-tuning as more data accumulates. Model improves with every student session.	Proprietary model that no competitor can replicate. This is the technical expression of the data flywheel moat.

6. AI Quality Control Framework (new in v4.2)

Six layers of quality control. Layers 1-3 are preventive -- they reduce the probability of a bad response before the LLM call. Layers 4-6 are reactive -- they detect and correct bad responses after they occur, and feed corrections back into the system.

6A. The six layers

Layer	Name	Mechanism	Phase	Cost
1	RAG grounding	Every prompt includes retrieved NCERT passages. LLM instructed: Answer using ONLY this content. Structural prevention of hallucination. If NCERT does not say it, the AI cannot say it.	1	Already built -- no extra cost
2	Structured prompt + temperature	System prompt enforces: step-by-step format, CBSE keywords, max 400 words, NCERT terminology. Temperature 0.3 for factual content. 0.7 only for analogies and examples.	1	Already built -- no extra cost
3	Student memory context	Knowing class level, subject, recent confusion, and weak topics narrows the answer space. Context-grounded responses are more accurate than context-free calls.	1	Already built -- no extra cost
4	Student signal loop	Thumbs up/down and Report incorrect on every response. helpful_rate tracked per subject and chapter. Below 65% triggers prompt review. This is the most important feedback signal.	1	Rs.0 per response -- student effort
5	Teacher review queue	Flagged responses appear at /admin/flagged. Teacher marks correct/wrong/partial. Corrections stored as labelled data. 8%+ correction rate on a topic triggers NCERT chunk audit.	1	Rs.0 per response -- teacher effort
6	Automated eval agent	Llama-8B (Gemma-2-9B) scores every response before showing: (1) NCERT-aligned? (2) Answers actual question? (3) Keywords present? Score below 6/10 triggers regeneration with a better model.	2	Rs.0.0025 per response (~Rs.0.75/student/month)

6B. Quality metrics to track from day one

Metric	How measured	Alert threshold	Action when triggered

helpful_rate per subject	Thumbs up / total responses in PostHog	Below 65%	Review and update doubtTutor.ts system prompt for that subject
flag_rate (incorrect)	Report button clicks / total responses	Above 5%	Immediate prompt and NCERT chunk audit for that chapter
regeneration_rate	Simpler mode + Exam answer chip taps / total	Above 40%	First response quality insufficient -- fix prompt depth and structure
teacher_correction_rate	Wrong marks / teacher-reviewed responses	Above 8%	NCERT chunks may be wrong or incomplete -- re-upload that chapter
avg_response_tokens	Token count logged per response	Above 600 tokens	Prompt not enforcing conciseness -- add explicit word limit instruction
worksheet_rejection_rate	Questions failing validation / total generated	Above 15%	Worksheet prompt quality issue -- review practiceGen.ts for that subject

6C. Quality improvement flywheel

Quality compounds over time through a structured data collection and correction loop.

Stage	What happens
Collect	Every response stored with student signal (helpful/flagged) and teacher correction if reviewed
Analyse	Monthly: identify lowest-performing subjects, chapters, and prompt versions. PostHog dashboard.
Improve prompts	Update system prompts for underperforming areas. A/B test new vs old. Track helpful_rate change.
Update NCERT chunks	High correction rate on a topic = chunk quality issue. Re-extract or manually curate better chunks.
Fine-tune (Phase 3)	Accumulated labelled data used to fine-tune Llama-8B. Better model on same NCERT content.

7. Agent Architecture (new in v4.2)

An agent is an LLM that can plan a sequence of actions, call tools, inspect results, and iterate until a task is complete. More powerful than a single LLM call. Also more expensive, slower, and more complex to debug. Use agents only where the extra loops justify the cost.

7A. Three agentic patterns -- use these before building full agents

These three patterns give 80% of agent value at 20% of the complexity. Implement these in Phase 1. Full autonomous agents in Phase 3.

Pattern	How it works	Padee use case
Tool use	LLM can call external functions: NCERT search, calculator, formula lookup, DB query. Returns results and incorporates them into response.	Doubt solver calls NCERT RAG retrieval as a tool. Math questions call a symbolic calculator for verification.
Multi-step (already built)	Two sequential LLM calls where the second has the context of the first. Analyse then Solve. DeepTutor dual-loop.	All text doubts: Call 1 identifies concept and misconception. Call 2 explains targeting that specific gap.
Self-check	Generate response -> pass to cheap model for validation -> regenerate if validation fails. The validator does not need to be smart, just accurate on a narrow check.	Worksheet generation: generate 15 questions -> Llama-8B validates each for NCERT accuracy -> regenerate failures.

7B. Task decision matrix -- agents vs simple calls

Task	Current approach	Use agent?	Reasoning	Phase
Simple CBSE doubt	Single call + RAG	No	NCERT has the answer. One call with good context is sufficient and faster.	Phase 1
Multi-step derivation	Dual-loop (already)	Agentic pattern	Two calls justified. Analyse first, then explain targeting the specific gap.	Phase 1
Worksheet generation	Single prompt, unvalidated	Add self-check now	Generate -> validate each question -> regenerate failures. Prevents wrong questions reaching teachers. One extra cheap call per question.	Phase 1
Photo doubt solving	Vision model + RAG	No	Vision model reads image, RAG provides context. Single call with good inputs produces accurate responses.	Phase 1

MCQ practice generation	Single JSON prompt	No	Llama-8B generates accurate structured MCQs on known NCERT topics. Add JSON validation but no agent loop.	Phase 1
Student diagnosis	Simple DB query on accuracy	Phase 2	Agent analyses 30+ sessions, identifies root misconception patterns across subjects. More accurate than surface DB query but adds cost.	Phase 2
Automated quality gate	Not built yet	Phase 2	Cheap LLM scores every response on 3 criteria before showing. Score below 6/10 triggers regeneration. Adds Rs.0.0025 per response.	Phase 2
JEE/NEET complex solving	Not in Phase 1	Full agent in Phase 3	Multi-step derivation requires: plan steps -> solve each -> check intermediate -> verify final. Errors in JEE solutions are high-trust violations.	Phase 3

7C. Worksheet validation agent -- Phase 1 implementation

The one agent pattern worth adding in Phase 1. A teacher who receives one factually incorrect question loses trust permanently. The validation step costs approximately Rs.0.08 per worksheet and prevents this.

```
// server/routes/ai.ts -- worksheet route (updated)

app.post('/api/ai/worksheet', async (c) => {
  const body = await c.req.json()
  const prompt = getWorksheetPrompt(body)

  // Step 1: Generate worksheet questions
  const raw = await llmComplete(MODEL.worksheet, [{role:'user',content:prompt}], 4096)
  const worksheet = JSON.parse(raw)

  // Step 2: Validate each question (self-check agent loop)
  const validated = []
  for (const section of worksheet.sections) {
    const checkedQuestions = []
    for (const q of section.questions) {
      const checkPrompt = `Is this question factually correct and NCERT-aligned for
        Class ${body.className} ${body.subject}?
        Question: ${q.question}
        Answer: ${q.answer}
        Reply with ONLY: VALID or INVALID: <reason>`
      const check = await llmComplete(MODEL.validation,
[{role:'user',content:checkPrompt}], 100)
      if (check.startsWith('VALID')) {
        checkedQuestions.push(q)
      } else {
        // Regenerate this question only
        const regenPrompt = `Generate one replacement question. ${check} -- fix this
issue.`
        const replacement = await llmComplete(MODEL.worksheet,
[{role:'user',content:regenPrompt}], 300)
        checkedQuestions.push(JSON.parse(replacement))
      }
    }
    validated.push({...section, questions: checkedQuestions})
```

```
  }

  return c.json({...worksheet, sections: validated})
})
```

Validation model: Llama-3.1-8B (Rs.0.004/1K tokens). At 15 questions per worksheet, average 100 tokens per validation call = Rs.0.006 per worksheet for validation. Negligible cost, high trust value.

7D. JEE/NEET complex solver agent -- Phase 3 design

Phase 3 introduces a full multi-step agent for JEE and NEET problem solving. This is the only Padee use case that genuinely requires autonomous agent loops.

Step	Agent action	Model
1 -- Classify	Identify problem type: derivation, numerical, conceptual, multi-concept. Determines tool set needed.	Llama-8B (cheap classifier)
2 -- Plan	Break problem into numbered sub-steps. Identify which formulae and concepts are needed.	Llama-70B or Claude
3 -- Solve step-by-step	Execute each sub-step. Call calculator tool for numerical steps. Call NCERT RAG for formula retrieval.	Claude Sonnet (best reasoning)
4 -- Self-check	Verify: Does each step follow from the previous? Does the final answer have correct units? Is it physically plausible?	Llama-70B (cheaper verification)
5 -- Format	Format as CBSE/JEE answer with step numbers, formula citations, and working shown.	GPT-4o-mini

Estimated cost per JEE complex problem: Rs.0.50-1.50 (4-5 LLM calls, mixed models). Justified for JEE students who pay Rs.1,500/year and for whom correct solutions are non-negotiable.

8. Phase 1 Build Scope (updated in v4.2)

Phase 1 now includes three quality control additions: student signal buttons, teacher review queue, and worksheet validation agent. The tech stack table is updated with quality monitoring tools.

8A. Phase 1 IN scope -- updated

IN Phase 1 (complete list)

* CBSE Class 8-12, all core subjects via NCERT RAG
* Ask AI -- text doubt (dual-loop, RAG, streaming)
* Ask AI -- photo doubt (Llama-3.2-11B Vision)
* Visual explain chip
* Persistent student memory injection
* Semantic response cache (pgvector)
* LiteLLM model routing
* MCQ practice engine
* Basic test mode + AI results insights
* Student dashboard -- all 10 modules
* XP + streak system (Supabase)
* Thumbs up/down signal on every AI response [NEW]
* Report incorrect answer button [NEW]
* Teacher review queue at /admin/flagged [NEW]
* Quality metrics in PostHog dashboard [NEW]
* Worksheet generator with validation agent [NEW]
* CBSE Paper Mimic (PDF upload)
* Teacher command centre
* Admin panel for NCERT PDF ingestion
* Login / signup (Supabase Auth)
* Daily usage cap (10 doubts/day free tier)
* Deployed to padee.ai with HTTPS
* PostHog + Sentry monitoring

OUT of Phase 1 (unchanged)

* IGCSE curriculum (Phase 2)
* JEE / NEET / CA Foundation tracks (Phase 3)
* Automated evaluation agent (Phase 2)
* Student diagnosis agent (Phase 2)
* JEE/NEET complex solver agent (Phase 3)
* React Native mobile app (Phase 2)
* Push notifications (Phase 2)
* Full test series (Phase 2)
* Human tutoring marketplace (Phase 3)
* Hindi language UI (Phase 4)
* Fine-tuning pipeline (Phase 3)
* NEET / JEE full mock tests (Phase 3)
* NEP compliance reports (Phase 4)
* White-label product (Phase 4)

8B. Updated tech stack for Phase 1

Layer	Technology	Notes

Frontend	React 18 + Vite 4 (existing prototype)	Do not rebuild. Wire backend only.
Backend	Hono + TypeScript (server/ folder)	Port 3001, proxied via Vite in dev.
LLM layer	LiteLLM	Any model, any provider. Config via .env.
Photo doubts	Llama-3.2-11B Vision via Groq [UPDATED]	Replaces GPT-4o-mini. 40% cheaper output tokens.
Vector store	Supabase pgvector	NCERT chunks + semantic response cache.
Embeddings	OpenAI text-embedding-3-small	Rs.0.0017/1K tokens. Full NCERT corpus: ~Rs.8 one-time.
Quality signals	PostHog [NEW]	helpful_rate, flag_rate, correction_rate tracked per subject.
Error monitoring	Sentry	Production error alerts.
Teacher review queue	Supabase DB + /admin route [NEW]	Flagged responses table. Teacher marks correct/wrong.
Database + Auth	Supabase PostgreSQL	Auth, DB, pgvector, storage all in one.
Frontend deploy	Vercel	padee.ai. Zero-config.
Backend deploy	Railway	api.padee.ai. Env vars in dashboard.

8C. Phase 1 success gate (unchanged)

Do not proceed to Phase 2 until all six are true. Behaviour, not opinion.

Metric	Threshold	Why
Weekly active students	500+	Minimum signal product is being used
Day 3 return rate	>40% unprompted	Core habit formation signal
Doubt resolution rating	>3.5 / 5 average	NCERT-grounded answers must be accurate
Teacher worksheet repeat	>60% generate 2+	Teacher adoption beyond novelty
Willingness to pay	>30% say yes to Rs.299/month	Business viability signal
Day 7 return rate	>25% unprompted	Strongest habit signal available

9. Four-Phase Roadmap (updated agent + quality milestones)

Phase 1 Months 1-4	Phase 2 Months 5-9	Phase 3 Months 10-15	Phase 4 Months 16-24
CBSE 8-12 + RAG + Quality signals + Worksheet validation agent + LiteLLM OSS routing	IGCSE + Foundation tracks + Automated quality gate agent + Student diagnosis agent + Mobile app	JEE/NEET/CA full + JEE agent solver + Fine-tuned Llama-8B model + Human tutoring + B2B	State boards + Hindi UI (Qwen-2.5-72B) + Offline mode + NEP reports + Scale to 50K+

10. Competitive Moats (updated with quality and agent moats)

Eight moats. The first six are unchanged from v4.1. Two new moats added relating to quality control infrastructure and agent architecture.

Moat	Description	Defensibility
1 Data flywheel	Every doubt, every answer rating, every teacher correction builds proprietary knowledge. After 12 months: Padee knows which NCERT lines confuse students most. No competitor can replicate this without the same students.	Compounds -- cannot be bought
2 NCERT accuracy	Answers from uploaded, current NCERT content. Syllabus updates: re-upload one PDF. Competitors on training memory are immediately stale when CBSE revises.	Updated faster than competitors
3 Lifecycle lock-in	Class 8 through competitive exam on one platform. 4 years of XP, streak, badges, AI context. Switching cost grows with every session.	Grows with time
4 Curriculum depth	CBSE + IGCSE + NEP + JEE + NEET + CA Foundation content across all classes. New entrant with better model but no curriculum is useless.	18 months to replicate
5 NEP relationships	Schools mandated to adopt NEP 2020. Platform embedded in NEP compliance reporting cannot be removed without administrative disruption.	Institutional switching cost
6 LLM-agnostic	Can switch to any better or cheaper model in one .env change. Competitors locked to a single provider face pricing risk and capability ceilings.	Permanent flexibility
7 Quality infrastructure [NEW]	Teacher correction data, student ratings, and automated quality scores form a quality dataset no competitor can acquire. This data directly drives the fine-tuning moat.	Converts to model moat
8 Fine-tuned model [NEW]	Phase 3 Llama-8B fine-tuned on 50,000+ labelled student sessions outperforms base GPT-4o on CBSE tasks at 10x lower cost. Proprietary model that cannot be replicated without the same data.	Technical + data moat combined

11. Design System (unchanged)

Option B -- Teal + Coral palette. Base font size 16px. Full spec in Padee_UI_Spec_v2.md.

Token	Value	Usage
--teal	#0D9488	Brand primary -- nav active, AI orb, progress rings
--coral	#EA580C	Main CTAs, challenge buttons, urgency actions
--navy	#0F1729	AI hero card background only
--amber	#D97706	Streak flame, at-risk alerts, weak topic cards
--bg-page	#F8F7F4	Warm white page background
Font base	DM Sans 16px	Increased from 14px. Fixes all small-text issues.

12. Success Metrics (updated with quality metrics)

Metric	Phase 1	Phase 2	Phase 3	Phase 4
Weekly active students	500+	5,000+	20,000+	50,000+
Day 7 retention	>25%	>35%	>40%	>45%
Doubt helpful_rate [NEW]	>65% per subject	>72%	>80%	>85%
Flag rate (incorrect) [NEW]	<5%	<3%	<2%	<1%
Cache hit rate	0% to 40%	50%+	60%+	65%+
AI cost/student/month	Rs.25 M1 -> Rs.15 M4	Rs.12	Rs.8 (fine-tuned model)	<Rs.6
NCERT accuracy (human eval)	>90%	>92%	>95% (fine-tuned)	>97%
School partnerships	3 pilot schools	25 schools	100 schools	500+ schools

13. Open Questions (updated)

1. Monetisation entry point: fully free to maximise student data for quality improvement, or freemium with 10-doubt cap to test willingness to pay early?
2. Worksheet validation agent false positive rate: at what rejection rate does the regeneration loop create teacher-visible delays? Need to test with real worksheets before setting threshold.
3. Fine-tuning trigger: what is the minimum labelled dataset size before fine-tuning Llama-8B produces measurable quality improvement over the base model on CBSE tasks? Likely 5,000-10,000 pairs but needs experiment.

4. Complexity routing threshold for JEE: how to detect whether a question is CBSE-level (route to Llama-70B) vs JEE Advanced (route to Claude)? Options: keyword detection, chapter metadata, or a lightweight 8B classifier call.
5. Teacher review queue incentive: should teachers be paid per correction reviewed, or is the value of accurate AI for their students sufficient motivation?
6. Demo mode scope for school presentations: Class 10 CBSE Physics student (teacher-focused) or Class 12 NEET student (parent-focused)?